

Q1. $A = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix}$

$$B = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix}$$

1. $2A - 3B = 2 \times \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix} - 3 \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 8 & -4 & 6 \\ 2 & 10 & -2 \\ 4 & 0 & 12 \end{bmatrix} + \begin{bmatrix} 3 & -12 & -6 \\ -9 & 6 & -15 \\ 0 & -21 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & -16 & 0 \\ -7 & 16 & -17 \\ 4 & -21 & 21 \end{bmatrix} \quad (\text{Ans})$$

2. $AB = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix} \times \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix}$

$$= \begin{bmatrix} -4 + 4 + 0 & 16 + 4 + 2 & 8 - 10 - 9 \\ -4 - 6 + 0 & 4 - 10 - 7 & 2 + 25 + 3 \\ -2 + 0 + 0 & 8 - 0 + 12 & 4 + 0 - 18 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & 22 & -11 \\ 12 & -13 & 30 \\ -2 & 20 & -14 \end{bmatrix} \quad (\text{Ans})$$

$BA = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix} \times \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix}$

$$= \begin{bmatrix} -4 + 4 + 4 & 2 + 20 + 0 & -3 - 4 + 12 \\ 12 - 2 + 10 & -6 - 10 + 0 & 9 + 2 + 0 \\ 0 + 7 - 6 & 0 + 25 + 0 & 0 - 7 - 18 \end{bmatrix} = \begin{bmatrix} 4 & 22 & 5 \\ 20 & -16 & 11 \\ 1 & 25 & -25 \end{bmatrix} \quad (\text{Ans})$$

$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$
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3. As we have already calculated AB and BA and observed that $AB \neq BA$; matrix multiplication is not commutative for given matrices.

$$4. A^T = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix}^T = \begin{bmatrix} 4 & 1 & 2 \\ -2 & 5 & 0 \\ 3 & -1 & 6 \end{bmatrix}$$

$$B^T = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix}^T = \begin{bmatrix} -1 & 3 & 0 \\ 4 & -2 & 7 \\ 2 & 5 & -3 \end{bmatrix}$$

$$5. \text{L.H.S.} = (AB)^T$$

$$= \begin{bmatrix} -10 & 41 & -11 \\ 14 & -13 & 30 \\ -2 & 50 & -14 \end{bmatrix}^T = \begin{bmatrix} -10 & 14 & -2 \\ 41 & -13 & 50 \\ -11 & 30 & -14 \end{bmatrix}$$

$$\text{R.H.S.} = B^T A^T$$

$$= \begin{bmatrix} -1 & 3 & 0 \\ 4 & -2 & 7 \\ 2 & 5 & -3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 2 \\ -2 & 5 & 0 \\ 3 & -1 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -4-6+0 & -1+15+0 & -2+0+0 \\ 16+4+21 & 4-10-7 & 8+0+12 \\ 8-10-9 & 2+25+3 & 4+0-18 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & 14 & -2 \\ 41 & -13 & 50 \\ -11 & 30 & -14 \end{bmatrix}$$

$$\therefore \text{L.H.S.} = \text{R.H.S.} \quad (\text{proved})$$

$$6. \operatorname{tr}(A) = 4 + 5 + 6 = 15$$

$$\operatorname{tr}(B) = -6$$

$$\operatorname{tr}(AB) = -37$$

$$\operatorname{tr}(BA) = -37$$

$$\therefore \operatorname{tr}(AB) = \operatorname{tr}(BA)$$

Q2.

$$7. C = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & -2 & 1 \end{bmatrix}$$

$$C + D = \begin{bmatrix} 3 & 3 & -1 \\ 2 & 7 & 6 \\ 3 & -4 & 6 \end{bmatrix}$$

$$C - D = \begin{bmatrix} 1 & -1 & -1 \\ 4 & 1 & -2 \\ -1 & 0 & 4 \end{bmatrix}$$

$$\begin{aligned} 8. \det(C) &= 2(20 + 4) - 1(15 - 2) - 1(-6 - 4) \\ &= 48 - 13 + 10 \\ &= 45 \end{aligned}$$

$\therefore C$ is not singular

$$9. C^{-1} = \frac{1}{|C|} \operatorname{adj} C$$

$$C_{11} = 24$$

$$C_{12} = -13$$

$$C_{21} = -7$$

$$C_{22} = 11$$

$$C_{31} = 15$$

$$C_{31} = 6$$

$$C_{32} = -7$$

$$C_{33} = 15$$

$$\operatorname{adj} A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$\therefore C^{-1} = \frac{1}{|C|} \begin{bmatrix} 24 & -13 & -10 \\ -7 & 11 & 5 \\ 6 & -3 & 5 \end{bmatrix}^T$$

$$= \frac{1}{45} \begin{bmatrix} 24 & -3 & 6 \\ -13 & 11 & -7 \\ -10 & 5 & 5 \end{bmatrix}$$

$$10. CC^{-1} = \frac{1}{45} \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix} \begin{bmatrix} 24 & -3 & 6 \\ -13 & 11 & -7 \\ -10 & 5 & 5 \end{bmatrix}$$

$$= \frac{1}{45} \begin{bmatrix} 48 - 13 + 10 & -6 + 11 - 5 & 12 - 7 - 5 \\ 72 - 52 - 20 & -9 + 44 + 10 & 18 - 28 + 10 \\ 24 + 26 - 50 & -3 - 22 + 25 & 6 + 14 + 25 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

$$11. \quad 2x + y - z = 7$$

$$3x + 4y + 2z = 16$$

$$x - 2y + 5z = 9$$

$$I. \quad CX = Y \quad \text{form,}$$

$$\begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 16 \\ 9 \end{bmatrix}$$

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The C matrix is ~~the same~~ identical to C in Q2.

$$\therefore C^{-1} = \frac{1}{45} \begin{bmatrix} 24 & -3 & 6 \\ -13 & 11 & -7 \\ -10 & 5 & 5 \end{bmatrix}$$

$$\therefore X = C^{-1}Y = \frac{1}{45} \begin{bmatrix} 24 & -3 & 6 \\ -13 & 11 & -7 \\ -10 & 5 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 16 \\ 9 \end{bmatrix}$$

$$= \frac{1}{45} \begin{bmatrix} 24 \times 7 - 3 \times 16 + 6 \times 9 \\ -13 \times 7 + 11 \times 16 - 7 \times 9 \\ -10 \times 7 + 5 \times 16 + 5 \times 9 \end{bmatrix}$$

$$= \frac{1}{45} \begin{bmatrix} 174 \\ 22 \\ 55 \end{bmatrix}$$

$$\therefore x = 174/45 ; y = 22/45 ; z = 55/45$$

$$12. \quad CD = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 1 + 2 & -4 + 3 + 2 & 0 + 4 - 1 \\ 3 - 4 + 4 & 6 + 12 - 4 & 0 + 16 + 2 \\ 1 + 2 + 10 & 2 - 6 - 10 & 0 - 8 + 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 3 \\ 3 & 14 & 18 \\ 13 & -14 & -3 \end{bmatrix}$$

Since matrix multiplication is row-by-column operation, the order matters. A changed order would mean that for each entry, a different row will be multiplied with a different column.